Morse code decoder

Ⅰ. About The Product

1.Parameters

1. Name: Morse code decoder
2. Working voltage: 9V
3. PCB length: 101*50mm, pure manual measurement, there is 1~2mm error

2.Application

1. Family Education
2. School Courses
3. Electronic Competition
4. DIY Production
5. Soldering Skills
6. Kit Sales
7. Gift Giving

3.Feature

1. Small size
2. Easy installation
3. Clear and clear working principle

Ⅱ. Important Notes

1.Necessary Tools

Soldering Irons, Solder Wire, Multimeters, Tweezers, Wire Cutters and other professional equipment

2.Installation instructions

This product is a DIY kit and you need to install it yourself.DIY installation is quite a delicate operation, you need to have patience to complete this project.

It is not a finished product!!!!

1. It is strongly recommended to read the installation manual before starting the installation.

2. Check the parameters and quantity of accessories against the accessories list.

3. Please refer to the accessories list and PCB screen printing to confirm the installation direction, position, positive and negative of the accessories.

4. Short circuits are strictly prohibited, which will damage the circuit components.

5. Ensure that each solder joint can be welded in good condition. If problems are found, check whether the solder joint falls off, virtual welding, and floating welding.

6. Do not let the soldering iron touch the component for too long, otherwise it will damage the component.

3.Service

1. Our main business is a variety of electronic components experimental practice kits and supporting tools, which are mainly used for training teaching, production courses, and welding exercises for the majority of electronic DIY enthusiasts. We are committed to allowing each customer to turn the purchased kit into a finished product, so that every assembly can be harvested. We insist on ensuring product quality with details and attitude, and improving service quality with professionalism and patience.

2. The assembly of the kit requires a certain amount of professional knowledge, and the degree of difficulty is also different. I hope you can try your best to communicate with us when you encounter difficulties in assembly. We will patiently assist you in completing the assembly.

3. The kit consists of multiple accessories, which we have repeatedly counted and tested before shipment. However, due to logistics reasons, the accessories you received are missing or the performance of the accessories is poor. Please contact us as soon as possible, and we will arrange a reissue for you immediately.

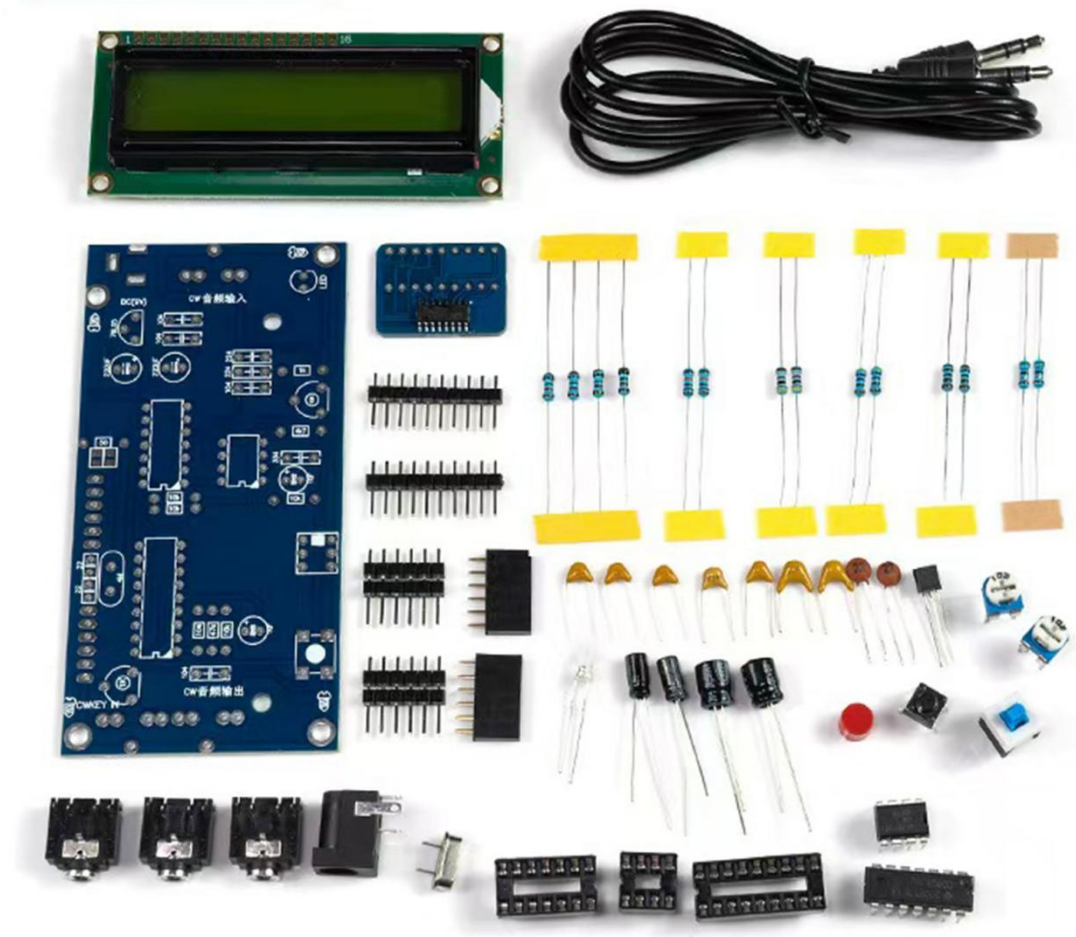
Ⅲ.Start Installation(Enjoy the installation process to the fullest! ! !)

1. Product Assembly Result

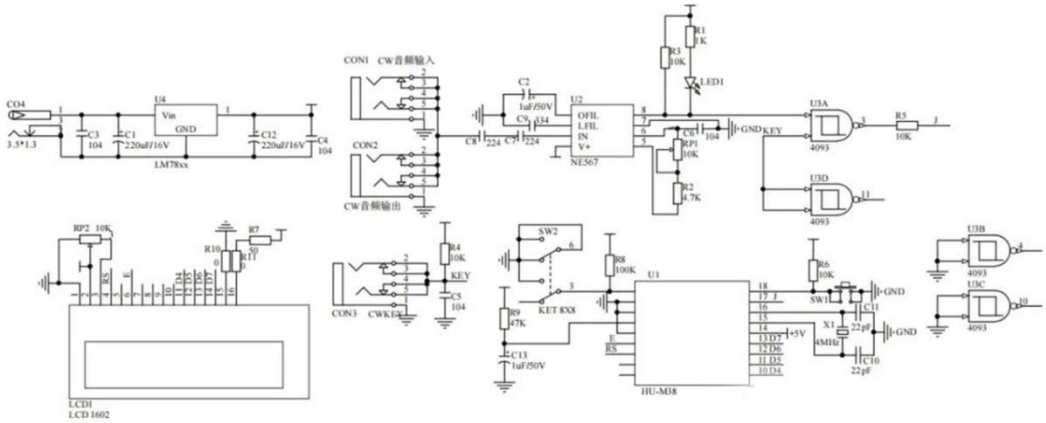
2. Component List

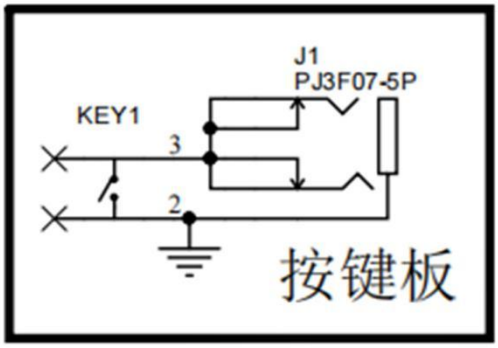
No.	Package	Name	Parameter	Label	QYT
1	Package A	Metal Film Resistors	50	50	1
2			1K	1K	1
3			4K7	4K7	1
4			10K	10K	4
5			47K	47K	1
6			100K	104	1
7	Package B	Ceramic Capacitors	22μF	22	2
8		Monolithic Capacitor	104	104	4
9			224	224	2
10			334	334	1
11		Electrolytic Capacitors	1μF	1μF	2
12			220μF	220μF	2
13		Triode	78L05	78L05	1
14		Crystal Oscillator	4MHz	4M	1
15		LED	3mm	LED	1
16		Potentiometer	102	1k	1
17			103	10k	1
18		Self-Locking Switches	8*8mm	/	1
19		Key Cap	/	/	1
20		Touch Switch	6*6mm	/	1
21		DC Power Socket	DC9V	DC(9V)	1
22		Audio Socket	/	CW音频输入; CW音频输出; CWKEY IN	3
23		Pin	9P	/	2
24			6P	/	2
25		Pin Header	6P	/	2
26	Package C	Touch Switch	12*12	KEY1	1
27		Key Cap	/	/	1
28		Audio Socket	/	J1	1
		Housing	/	/	2
29		Brass Pillar	M3*5+5	/	4
30		Screws	M3*5	/	4
31			M2*20	/	4
32		Nuts	M3	/	4
33			M2	/	4
34	Package D	Adapter Plate	25*20	/	1
35		PCB	50*100mm	/	1
36	Package E	IC	NE567	/	1
37			CD4093	/	1
38		IC holders	8P	/	1
39			14P	/	1
40			18P	/	1
41		liquid crystal screen	/	/	1
42		Audio cable	/	/	1
43		Power Supply	9V	/	1
44	Package F	Housing	/	/	2
45		Brass Pillar	M3*20	/	4
46			M3*5+5	/	4
47		Screws	M3*5	/	8

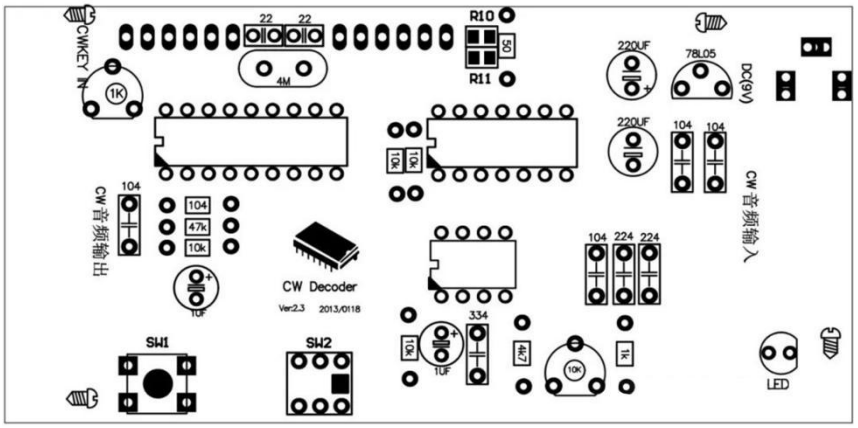
3.Product parts diagram

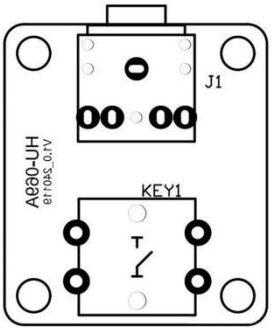


4. Circuit Principle

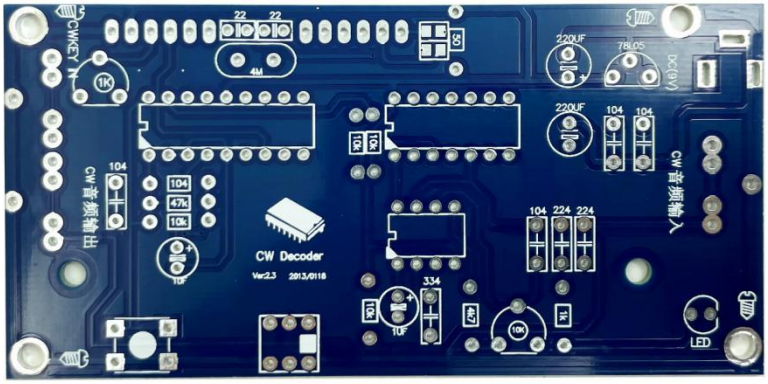








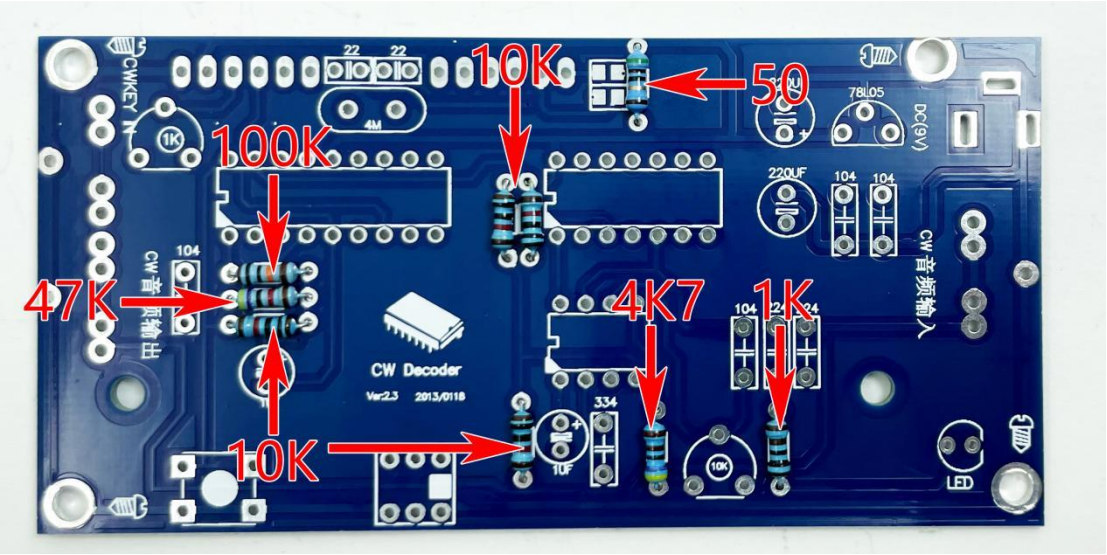
5. Soldering Steps



- **Step 1: Soldering Metal Film Resistors**

Name	Parameter	Label	QYT
Metal Film Resistors	50	50	1
	1K	1K	1
	4K7	4K7	1
	10K	10K	4
	47K	47K	1
	100K	104	1

Tips: Soldering metal film resistors does not require distinguishing positive and negative poles. It is recommended to use a multimeter to test the resistance value before installation.

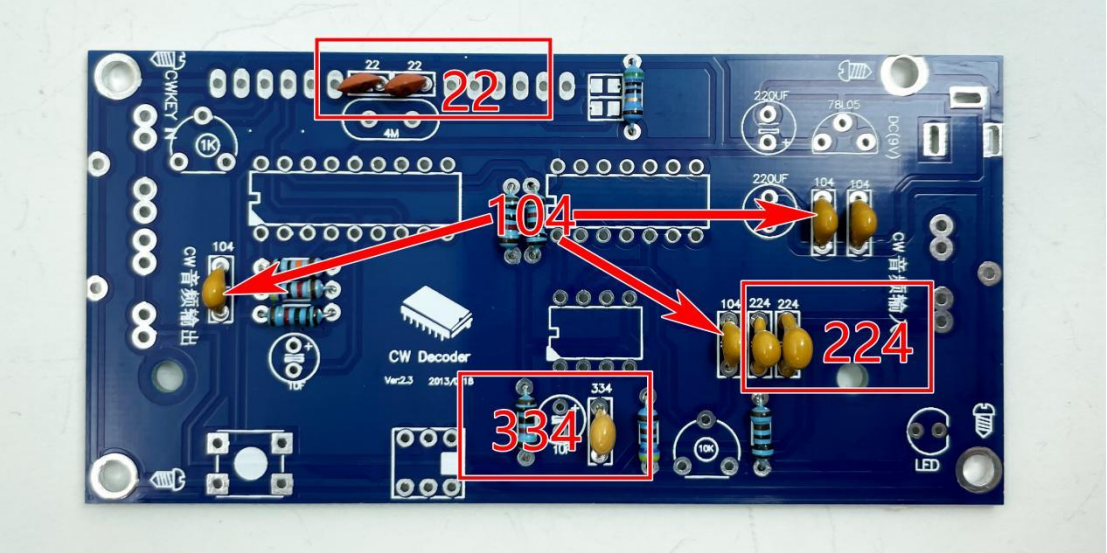


- **Step 2: Soldering Porcelain Chip Capacitors ; Soldering Monolithic Capacitors**

Name	Parameter	Label	QYT
Ceramic Capacitors	22μF	22	2
Monolithic Capacitor	104	104	4
	224	224	2
	334	334	1

Tips: Soldering porcelain chip capacitors without distinguishing between positive and negative terminals.

Soldered monolithic capacitors do not require distinguishing between positive and negative electrodes.

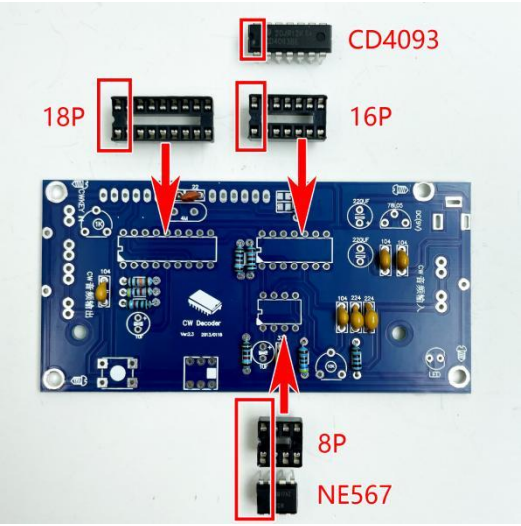


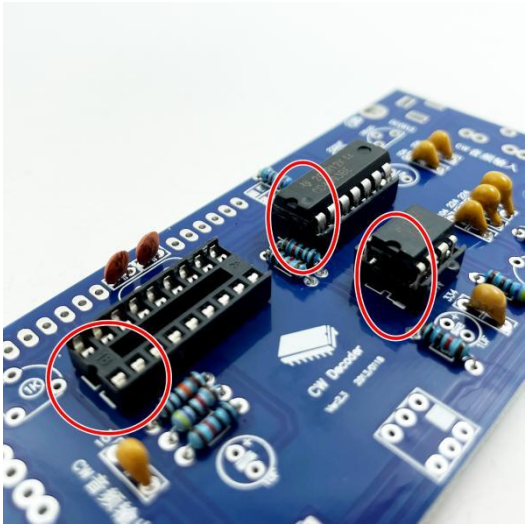
- **Step 3: Soldering IC**

Name	Parameter	Label	QYT
IC	NE567	/	1
	CD4093	/	1
IC holders	8P	/	1
	14P	/	1
	18P	/	1

Tips: 1、The half-circle notch on the IC holder corresponds to the half-circle notch on the screen-printed pattern for mounting, and the IC holder is soldered first, then the IC is mounted.

2、The half-circle notch on the IC corresponds to the half-circle notch on the screen printing pattern for mounting.

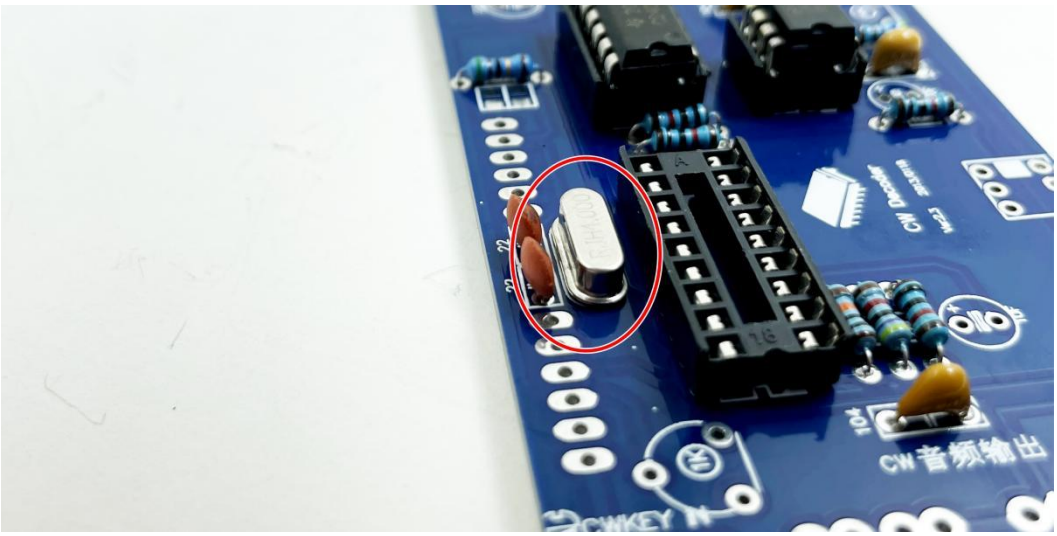




- **Step 4: Soldering Crystal Oscillator**

Name	Parameter	Label	QYT
Crystal Oscillator	4MHz	4M	1

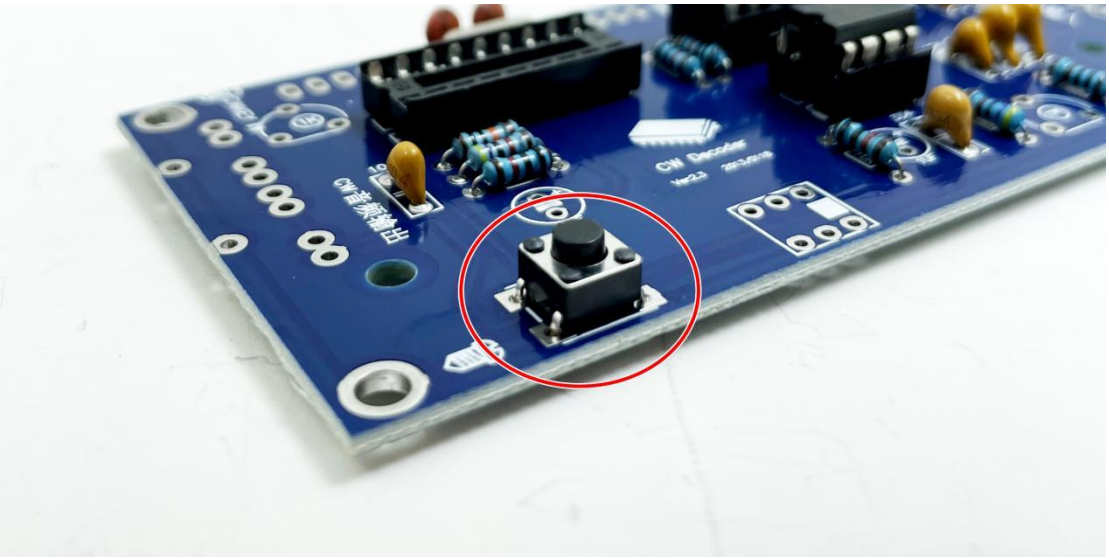
Tips: Soldering crystal oscillators do not require distinguishing between positive and negative electrodes.



- **Step 5:Soldering light touch Switches**

Name	Parameter	Label	QYT
Touch Switch	6*6mm	/	1

Tips: The pin on the flick switch corresponds to the through-hole on the PCB for mounting.



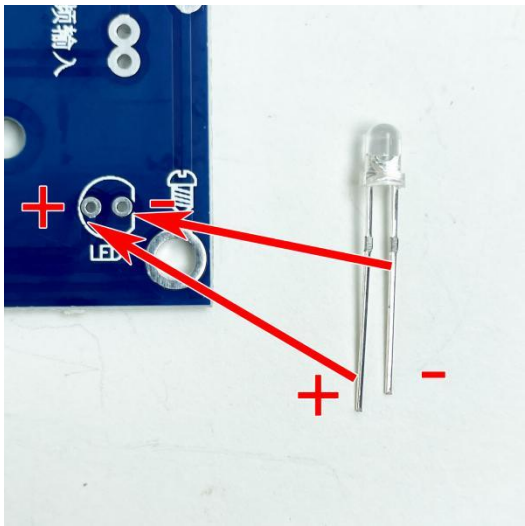
- **Step 6: Soldering LED**

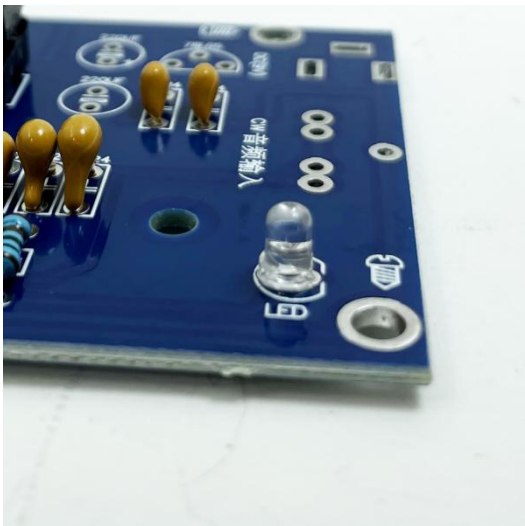
Name	Parameter	Label	QYT
LED	3mm	LED	1

Tips: The long pin of LED is positive, the short pin is negative (long is positive, short is negative).

The solder joint on the PCB near the end of the notch patterned with the brush is negative.

ps. On the bottom of the LED, there will be a notch of the same kind, and the pin on the LED near the notched end will be negative.

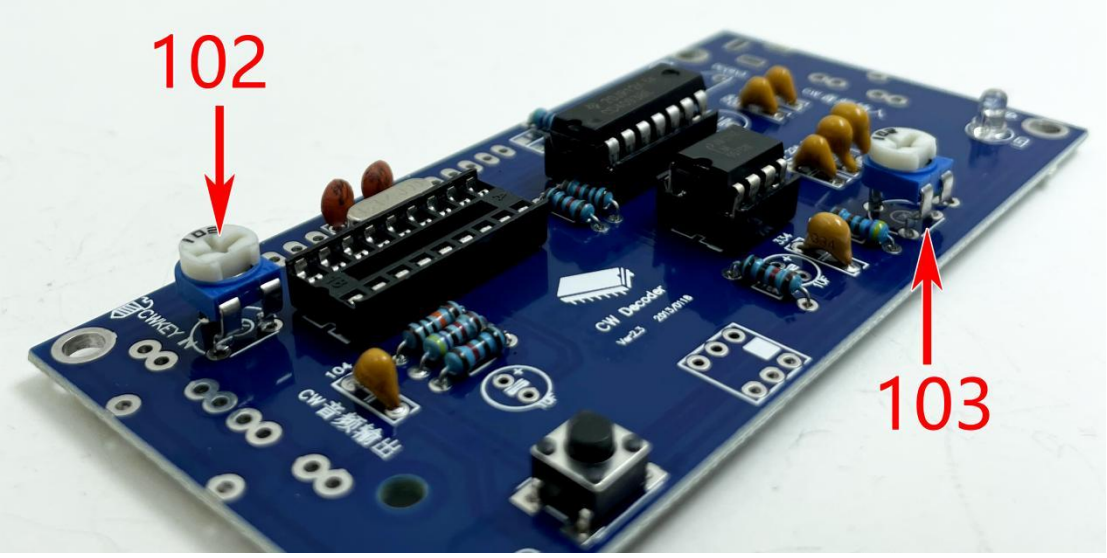




- **Step 7:Soldering Potentiometer**

Name	Parameter	Label	QYT
Potentiometer	102	1k	1
	103	10k	1

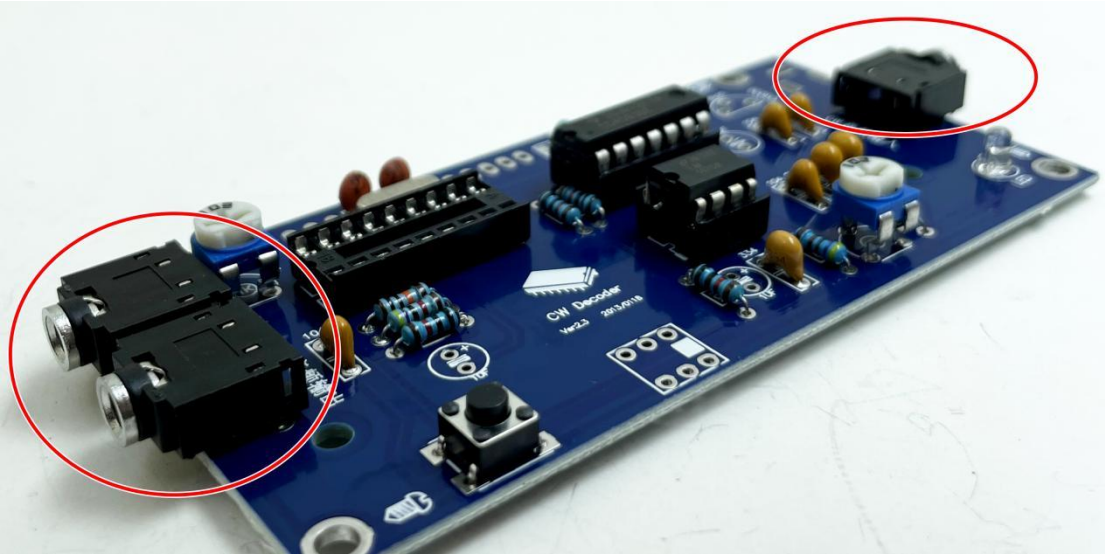
Tips: Install the potentiometer by matching the pins on the potentiometer to the through-holes on the PCB.



- **Step 8:Soldering Audio Sockets**

Name	Parameter	Label	QYT
Audio Socket	/	CW 音频输入;	1
Audio Socket	/	CW 音频输出;	1
Audio Socket	/	CWKEY IN;	1

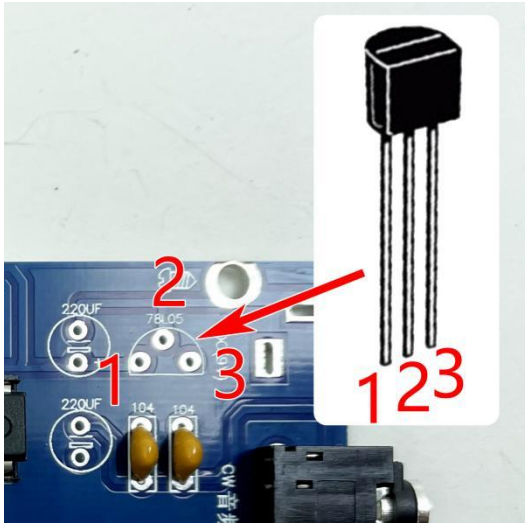
Tips: Install the audio socket by matching the pins of the audio socket to the through-holes on the PCB, paying attention to the direction of installation.

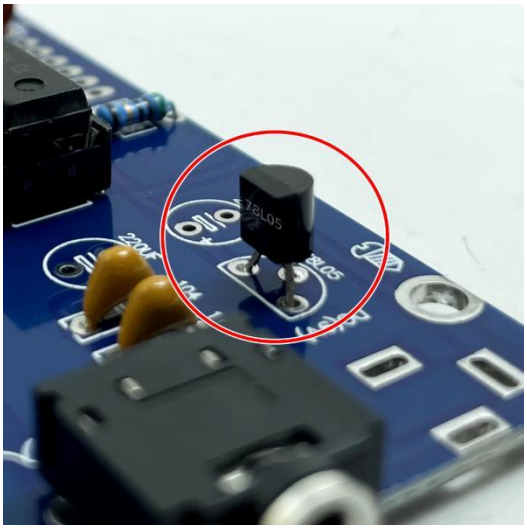


- **Step 9:Soldering Triode**

Name	Parameter	Label	QYT
Triode	78L05	78L05	1

Tips: The side of the transistor plane corresponds to the side of the circuit board printed in a straight line.





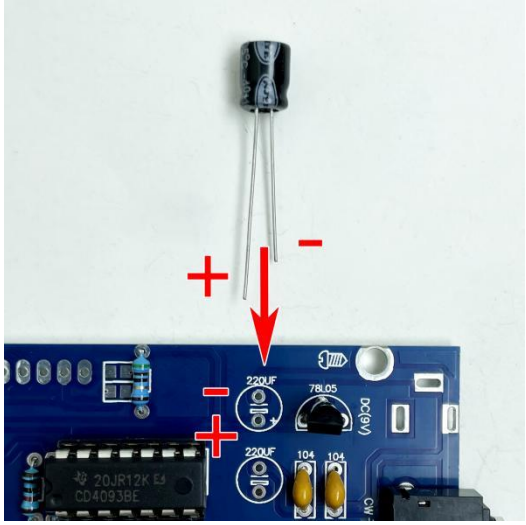
- **Step 10:Soldering Electrolytic Capacitors**

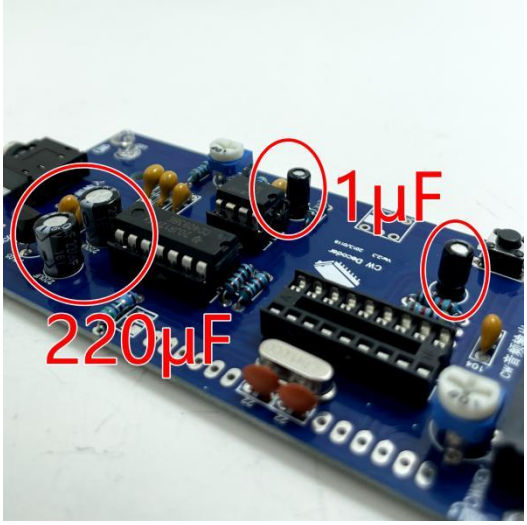
Name	Parameter	Label	QYT
Electrolytic Capacitors	1μF	1μF	2
	220μF	220μF	2

Tips: 1、 The long pin of the Electrolytic Capacitors is positive and the short pin is negative (long positive, short negative).

2、The pin on one end of the printed white strip on the electrolytic capacitor is negative and the pin on the other end is positive.

3、 The through-hole at the ""+"" end of the printed circuit board is positive, and the through-hole at the other end is negative.

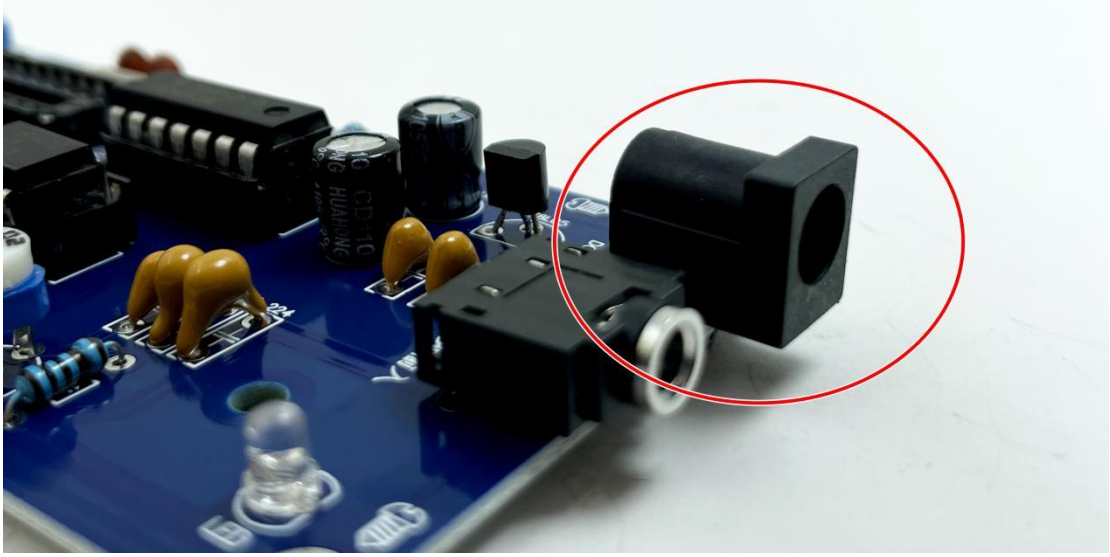




- **Step 11:Soldering DC Power Socket**

Name	Parameter	Label	QYT
DC Power Socket	DC9V	DC(9V)	1

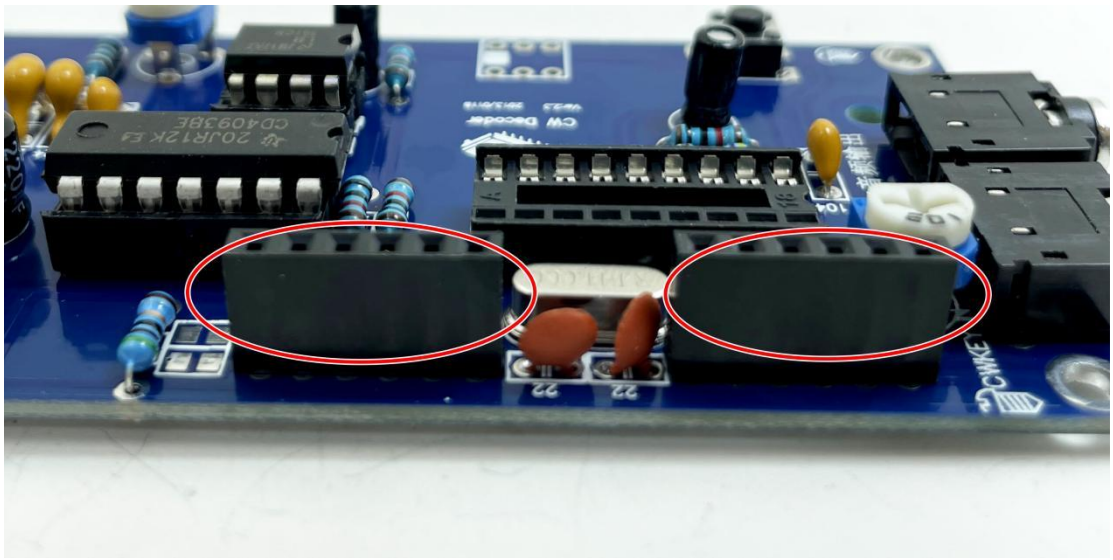
Tips: The pins on the DC power supply holder correspond to the through-holes on the PCB for mounting.



- **Step 12:Soldered Pin Headers**

Name	Parameter	Label	QYT
Pin Header	6P	/	2

Tips: Installation with reference to the prompted diagram.

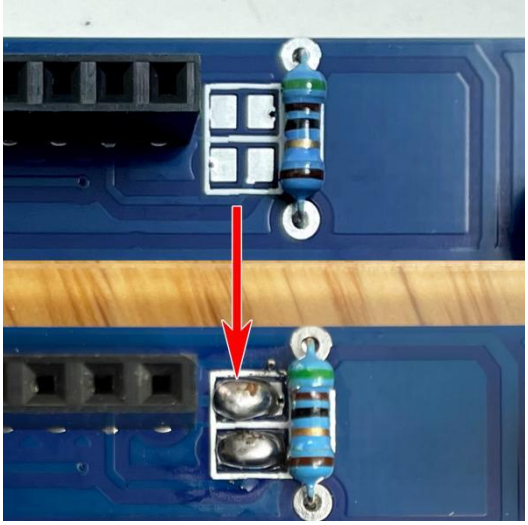


- **Step 13:Soldering Self-Locking Switches**

Name	Parameter	Label	QYT
Pin Header	16P	LCD1	1

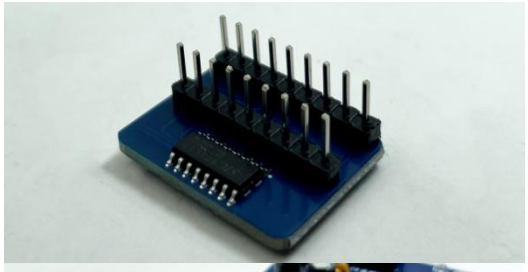
Tips: The marking on the self-locking switch corresponds to the marking on the screen-printed pattern for installation.

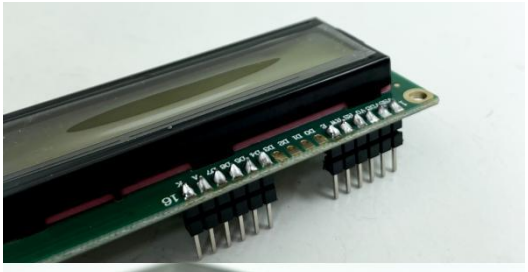
Short-circuit soldering of solder joints.Short-circuit solder the two solder joints as shown.

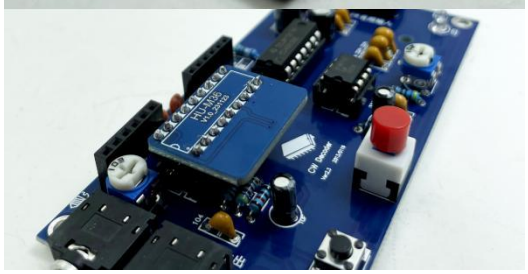


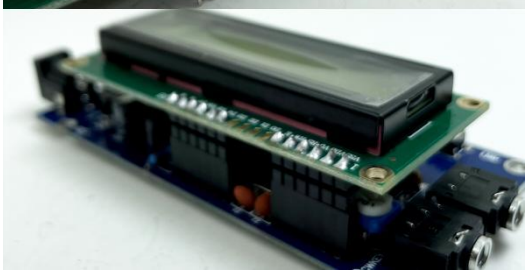
- **Step 14:Soldering Pins**

*Tips:Install according to the prompt diagram.*Be careful not to use brute force during installation to avoid damage to the pins.







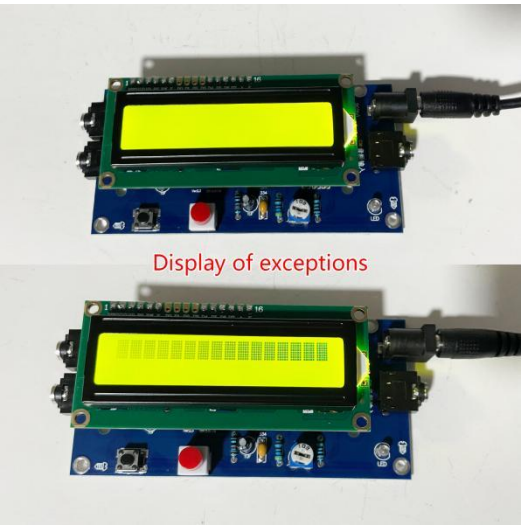


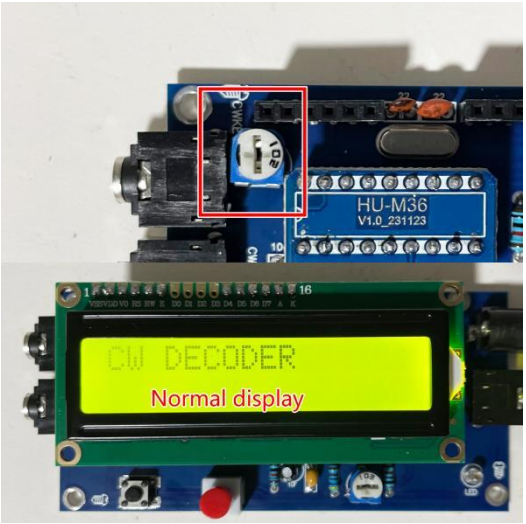
- **Step 15:Soldering light touch Switches and Audio Sockets**

  Tips: Mount the foot pins corresponding to the through-holes on the PCB, taking care to distinguish the direction.

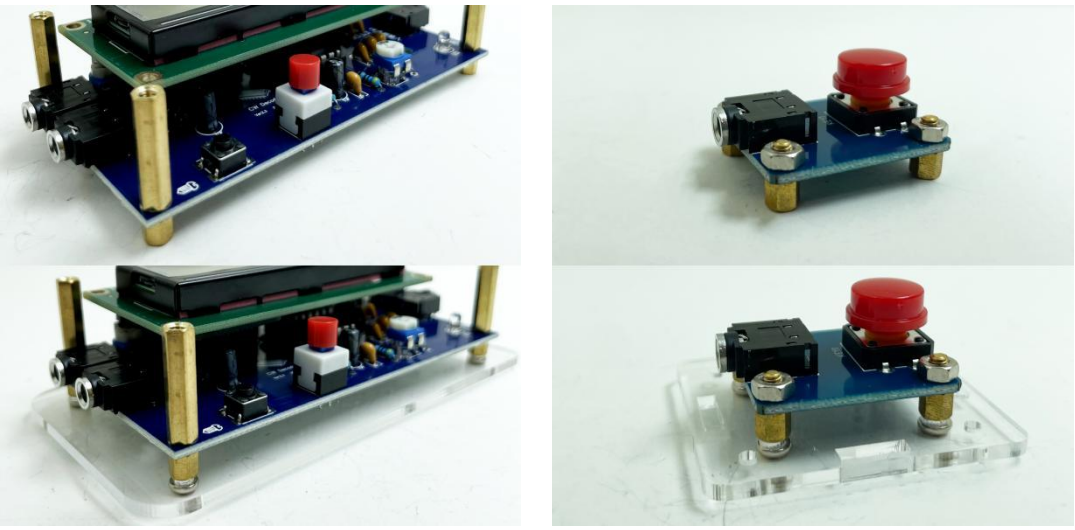
- **Step 16:Display Adjustment Gray Scale**

  Tips:Plug in 9v power to the display section, at this time the display will not light up or light up with no text, then you need to adjust the grayscale. Disconnect the power supply, remove the display and adjust the 102 potentiometer counterclockwise with a Phillips screwdriver. Repeat the adjustment until the display shows normal.

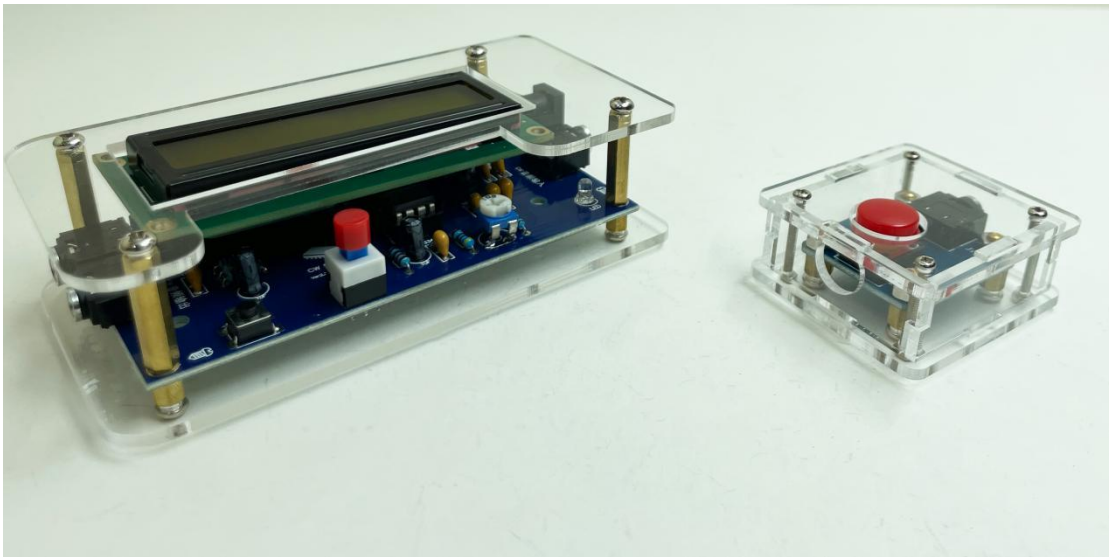




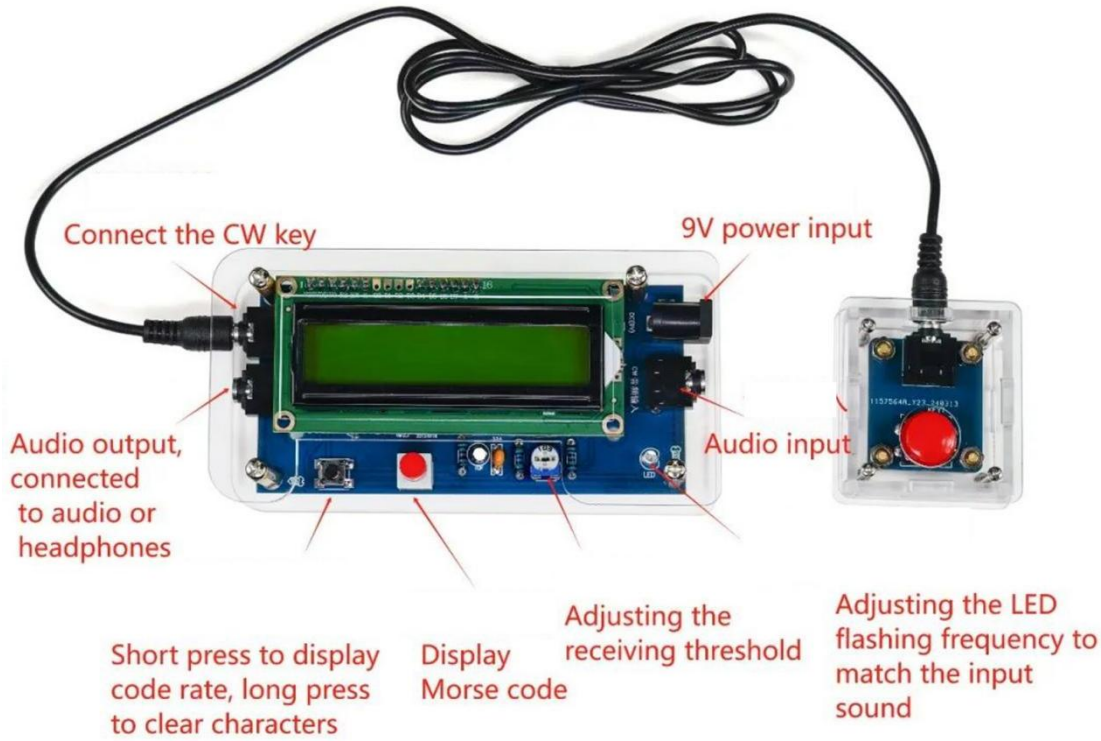
- **Step 17: Mounting brass posts and bottoms**

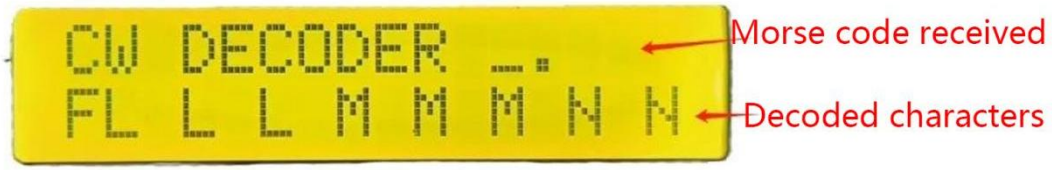


- **Step 18:Mounting Enclosure**



Ⅳ. Instructions for Use





(1)Connect to a DC9V power supply.

(2)The CW audio input terminal is connected to Morse code audio, which can be decoded, converted into characters, and displayed on the LCD screen.

(3)The CW audio output terminal can output the received Morse code audio, and can be connected to a speaker or headphones.

(4)Adjust the receiving threshold potentiometer to match the LED flashing frequency with the sound heard. (If not properly adjusted, the decoder cannot decode the audio)

(5)Press the self-locking switch to view the input Morse code, which will not be displayed when it pops up.

(6)Press the light touch switch to view the received Morse code rate.

(7)The CWKEY end is connected to the button board for practicing Morse code, and the Morse code entered by the button will be displayed on the LCD screen. (Cannot input at the same time as audio input, otherwise it will interfere with each other and cannot be decoded correctly)

(8)Short press the button board as a point, long press the button board as a line. (Long press the button for more than 360ms as a line, otherwise short press as a point)

(9)Morse code table

Letter

Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol
A	. ━	B	━. . .	C	━. ━.	D	━. .
E	.	F	. . ━.	G	━.	H	. . . .
I	. .	J	. ━━	K	━. ━	L	. ━. .
M	━	N	━.	O	━━	P	. ━.
Q	━. ━	R	. ━.	S	. . .	T	━
U	. . ━	V	. . . ━	W	. ━	X	━. . ━
Y	━. ━	Z	━. .				

Number

Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol
0	———	1	. ——	2	. . —	3	. . . —
4	. . . . -	5	. . . . .	6	-. . . .	7	—. . .
8	—. .	9	——.				

Punctuation marks

Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol	Character	Morse code symbol
?	. . —. .	,	—. . —	.	. -. -. -	!	-. -. —
@	. —. -.	:	—. . .	-	-. . . . -		

(10)Morse code practice examples

I			
I			
. .			
LOVE			
L	O	V	E
. —. .	— — —	. . . —	.
LIKE			
L	I	K	E
. —. .	. .	—. —	.
YOU			
Y	O	U	
—. ——	— — —	. . —	
GOOD			
G	O	O	D
——.	— — —	— — —	—. .
OK			

O	K		
— — —	—. —		
EAT			
E	A	T	
.	. —	—	